

Assignment 1

Question 1 (10 marks)

(a) Prove using Venn diagrams that: $A \setminus B = A \cap \overline{B}$.

(b) Prove using set identities that: $(C \setminus B) \setminus (B \setminus A) = C \setminus B$.

Note: The only identities you can use here are the laws of Boolean algebra, and the result of part (a).

Question 2 (10 marks)

Use double-inclusion to prove the following set identity.

$$(A \cap B) \times C = (A \times C) \cap (B \times C)$$

Question 3 (10 marks)

A website asks you to login in order to be able to write on its *guestbook*. But of course, it won't let you choose just any string as a password: that would be too simple! Instead, the requirements are the following:

1. The password must contain between 8 and 10 characters.
2. Each character must be of one of the following types: lowercase letters without accents (a–z), uppercase letters without accents (A–Z), digits (0–9) or *special characters* from the following list: !?#&@%*~\$ _
3. The password contains at least one character of at least three of the four types mentioned above.

In how many ways can a password be chosen in order to meet all those requirements?

Note: That way of thinking about internet security is completely obsolete, but somehow still common. XKCD has a comic strip about that... See <https://xkcd.com/936/>

Question 4 (10 marks)

How many integer solutions are there to the equation: $x_1 + x_2 + x_3 + x_4 = 20$

(a) If we only require $x_i \geq 0$ for all i ?

Hint: Stars and bars...

(b) If we require $0 \leq x_i \leq 9$ for all i ?

Hint: $x_i > 9$ if and only if $x_i = 10 + t$ for some integer $t \geq 0$...